Avishek Shrabon

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RESEARCH

HSBN-FL: Hierarchical Stochastic Bottleneck Networks for Heterogeneous Federated Learning

2025 — Present

Independent Research — In Progress, Core Experiments Complete

- Addresses a cross-disciplinary challenge at the intersection of **distributed systems, privacy, and machine learning**: enabling heterogeneous institutions (hospitals, NGOs, research labs) to collaboratively train models without sharing raw data or model weights
- Proposes replacing weight exchange with **latent-space communication** — clients share compressed representations rather than parameters, natively supporting architectural heterogeneity across participants
- Server structured as a two-level hierarchy: a **Transformer adapter (Z1)** attending over all client representations simultaneously, and an **MLP apex (Z2)** compressing the pooled output into a global abstract state; top-down feedback flows back to each client as a soft alignment signal
- Evaluated on CIFAR-100 across 20 heterogeneous clients under Non-IID Dirichlet partitioning; achieved **under 0.7% accuracy variance** across heterogeneity levels

Hierarchical Stochastic Bottleneck Networks: Conditions for Representational Abstraction in Deep Hierarchies

2025

Manuscript Under External Review — Targeting IEEE Conference

- Investigates under what conditions deep networks form a **genuine hierarchy of abstractions** rather than encoding redundant information across layers — a question spanning information theory, neuroscience, and deep learning
- Identifies three necessary mechanisms: per-level objectives, bandwidth-limited stochastic channels (KL penalty), and bidirectional message passing — each level learns a strictly more compressed representation than the one below
- Verified empirically on CIFAR-10 and CIFAR-100 using **Centered Kernel Alignment** and intrinsic dimensionality; ablations isolate two qualitatively distinct failure modes
- Manuscript complete preparing for submission to a lab and finally a IEEE conference

Adaptive Federated Learning with Heterogeneous Data and Client Distributions

2024 — 2025

Undergraduate Thesis — AIUB Academic Repository, Grade: A+

- Proposed a pipeline combining **gradient-based client clustering** with cross-architecture Knowledge Distillation (ResNet and ViT) to address non-IID data distributions in federated settings; taught myself the domain independently in approximately six weeks
- Supervised by **Debajyoti Karmaker**, formerly AIUB, now Lecturer at RMIT & AIH, Australia

EDUCATION

Bachelor of Science in Computer Science & Engineering

Jan 2022 — Dec 2025

American International University Bangladesh (AIUB) — Dhaka, Bangladesh

CGPA 3.74 / 4.00

- Completed degree in **3.5 years** — one semester ahead of standard duration
- Awarded the **Dean's Award** (Spring 2022–23) for academic excellence — CGPA above 3.8
- Notable coursework: Computational Statistics & Probability (A+), Algorithms (A), Software Requirements Engineering (A+), Basic Mechanical Engineering (A+)

EXPERIENCE

Private Tutor — Sciences & Mathematics

Mid 2022 — Present

Self-employed — Dhaka, Bangladesh

- Tutored 3–4 students simultaneously across Grade 8–12 in Physics, Chemistry, Biology, Mathematics, Higher Mathematics, and ICT — sustained alongside a full-time degree and independent research
- Developed strong ability to adapt explanation strategies to individual learners; regularly communicated complex ideas across different levels of prior knowledge

PROJECTS

1000 Missionary Movement Bangladesh — Training Platform

2025 — Present

Full-Stack Web Application — Live Deployment, Nationwide NGO

- Collaborated with NGO leadership and programme coordinators to translate a manual, paper-based intake process into a digital platform — required sustained cross-functional communication with non-technical stakeholders
- Built on **Next.js, Prisma, and Auth.js**; handles applicant intake, role-based access, and programme administration across hundreds of participants nationwide

Church Event Ledger & Ticket Reconciliation System

2025

Full-Stack Software — Live Deployment, 5 Consecutive Events

- Identified an operational bottleneck through direct observation of event logistics; built and deployed a working solution in 3 days in coordination with the event team
- Reduced end-of-night accounting time from 15–20 minutes to under 5 minutes — a **75% reduction**; adopted as standard practice across subsequent events

Proximity Sensor Aid for the Visually Impaired

2023

Hardware — Embedded Systems & Assistive Technology

- Bridged electrical engineering and human-centred design to build a wearable obstacle detection system using Arduino Nano, HC-SR04 ultrasonic sensor, and experimental Time-of-Flight sensor; critically evaluated limitations and proposed improvement path

Darkness-Triggered Street Light Automation

2023

Hardware — Electronics & Infrastructure

- Designed an LDR-based automatic street lighting system triggering relay-controlled lights based on ambient darkness — motivated by energy efficiency and public infrastructure improvement

SKILLS

LANGUAGES	Python, JavaScript, TypeScript, SQL, C++ (surface), C# (surface)
AI / ML	PyTorch, scikit-learn, Hugging Face, NumPy, Pandas, Seaborn, Matplotlib
WEB & FRAMEWORKS	React, Next.js, Nest.js, Node.js, Vite, HTML, CSS/SCSS
DATABASES	PostgreSQL, MongoDB, SQLite
TOOLS	Git, GitHub, VS Code, Google Colab, Kaggle, Linux
RESEARCH	Federated Learning, Knowledge Distillation, Distributed Systems, Information Theory, LLMs, NLP

AWARDS & ACTIVITIES

Dean's Award

Spring 2022-23

American International University Bangladesh

Runners-Up — Annual Volleyball Tournament

2024

American International University Bangladesh — competitive team sport

Volunteer — Church Community Services

Ongoing

Event coordination, software deployment, and community engagement — ongoing

REFERENCES

Anjir Ahmed Chowdhury

PhD Researcher, Intelligent Data and Systems Lab — University of Houston, USA

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Debajyoti Karmaker

Lecturer (RMIT & AIH), Australia — Undergraduate Thesis Supervisor

Further contact details available upon request.